

**VIETNAM NATIONAL UNIVERSITY – HO CHI MINH CITY
THE INTERNATIONAL UNIVERSITY**

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING



**WEB APPLICATION DEVELOPMENT LAB
IT093IU - Group 02**

PROJECT REPORT

Topic: Food Recipe Website

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CHAPTER 1: INTRODUCTION

1.1 Abstract:

This web application development project involves building an online Food Recipe Website. The project is developed using Java for the back-end and MySQL for database management, with GitHub support to track the project's progress and enable collaboration among team members. The project also emphasizes storing information about. The database structure is carefully designed to maintain system scalability and efficiency, supporting smooth operation across all processes. Furthermore, the design process defines data entities, relationships, attributes, and constraints logically to maintain data integrity. The user interface is designed to be intuitive, enabling users to check for food categories, search for specific food recipes, and add or edit recipes, while administrators can manage users, check statistics and generate reports.

1.2 Project Overview:

A Food Recipe Website allows users to register, log in and add recipes with different access levels (user, administrator, editor). Users can search for food recipes, select a desired recipe. Administrators can manage the recipes data. All data including user information, recipe details will be stored and managed using MySQL. The editors can edit the recipes. The database is supported with Entity Relationship Diagrams (ERDs) and schema diagrams to provide a clear visualization of entities and relationships, ensuring efficient SQL queries and reliable data operations.

1.3 Tool & Environment:

This project utilized the following tools and environment:

HTML/CSS: Website main framework and format the HTML webpage in a unique-looking way.

JS: for backend development to handle logic and recipe details.

Draw.io: Design ER Diagram and Relational Schema.

1.4 Installation & Setup:

Clone the repository: `git clone `HeomapTV123/Recipes.github.io``

To set up the project, run “run npm dotenv” in the backend folder to access the session token and port to host the server. The configuration can be changed as desired, but this project is built with “Port: 5000”.

To get the website running, open a browser and put “localhost:5000/html/main.html” to access the main homepage.

CHAPTER 2: METHODOLOGY

2.1 Requirement Analysis:

This project is a recipe management platform that allows users to create, manage, categorize, and interact with cooking recipes. The system supports user accounts, recipe publishing, ingredients, step-by-step instructions, tagging, categorization, comments, ratings, and favorites.

2.2 Approach Analysis:

2.2.1 Review Materials:

- NodeJS
- HTML, CSS, JavaScript, mySQL

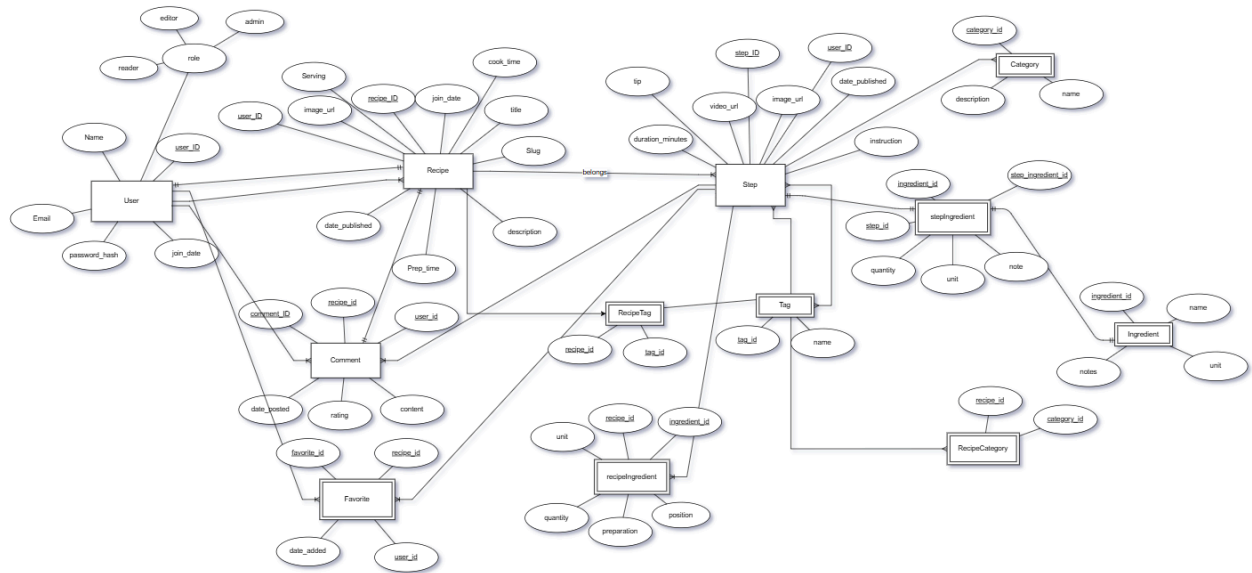
2.2.2 Research Approach:

- Learning Syntax of required programming languages (HTML, CSS, JavaScript, MySQL, NodeJS)

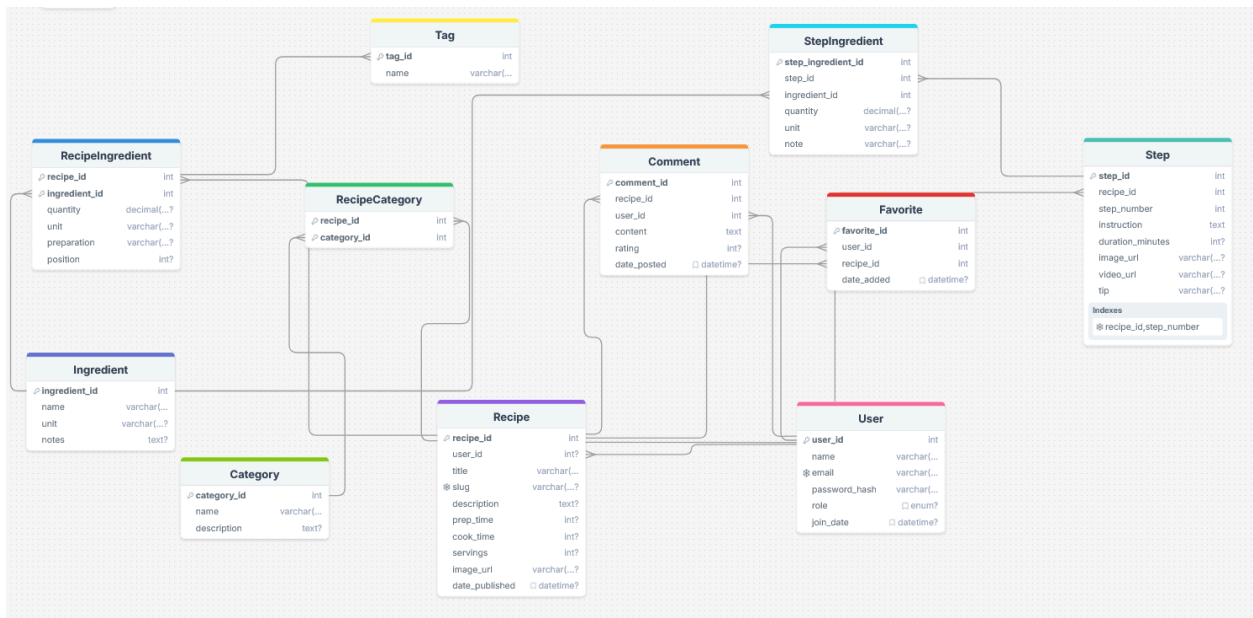
2.2.3 System Analysis:

- Strategic Choices in System Development:
 - + Front-end: HTML, CSS, JavaScript
 - + Back-end: NodeJS
 - + Database: MySQL

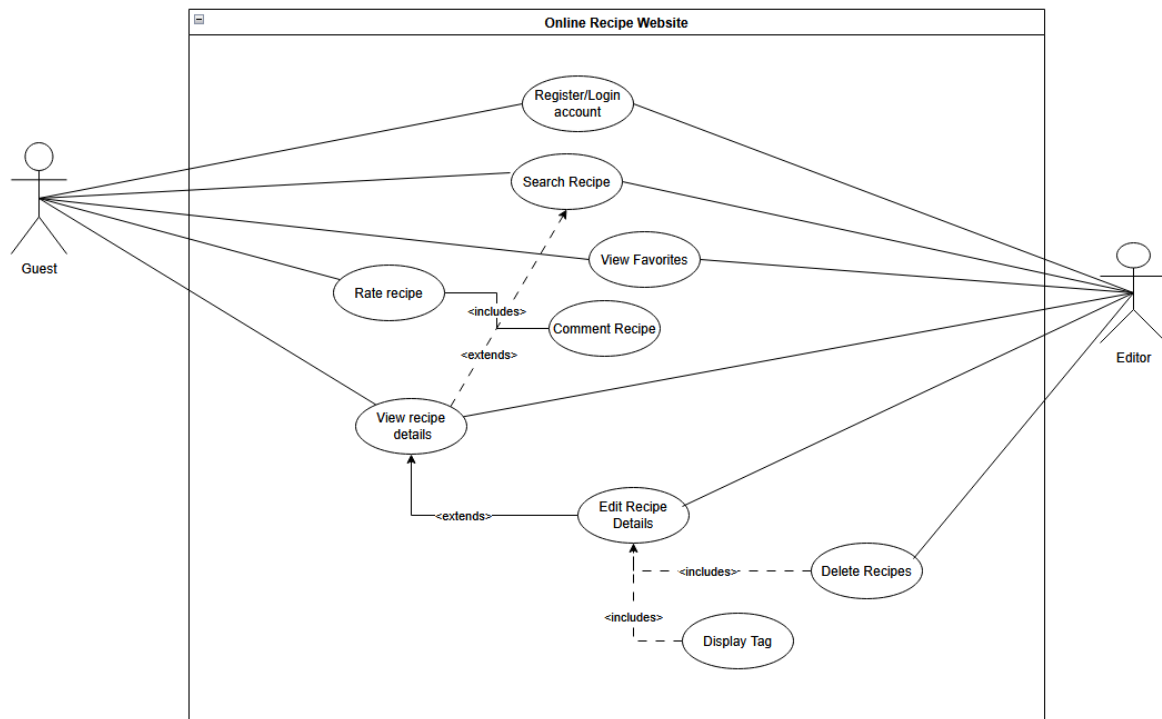
2.3 ERD:



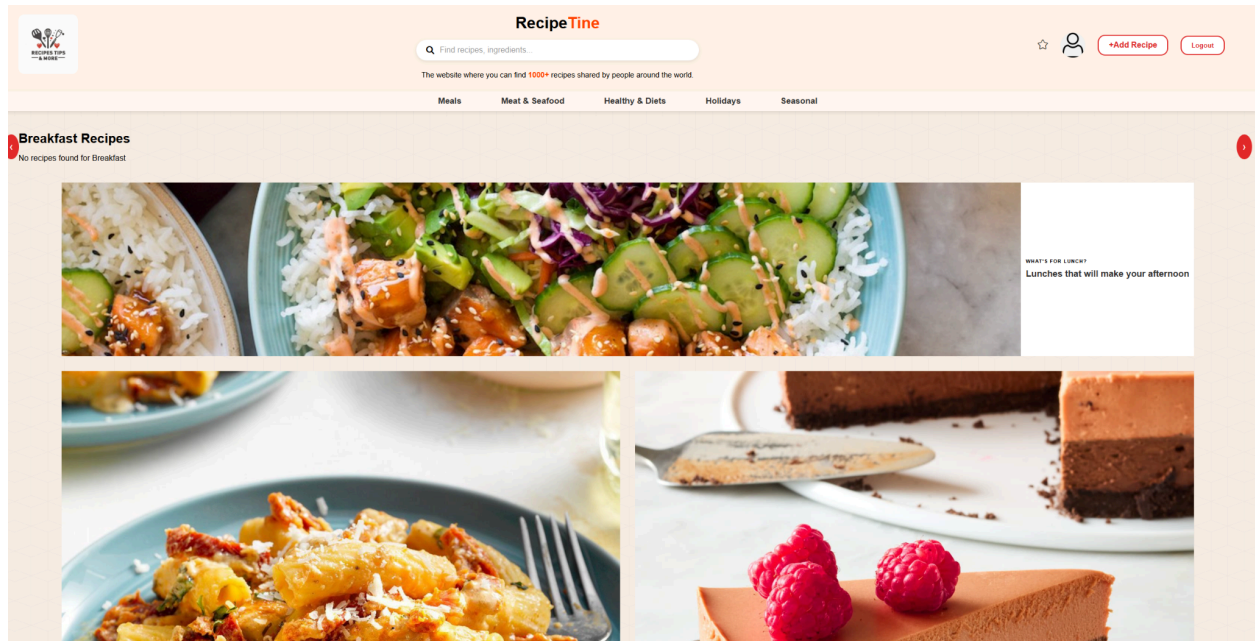
2.4 Schematic:



2.5 Use Case Diagram:



2.6 Demonstration:



Figures 4.1: Homepage

Add Your Own Recipe!

Recipe Name

Category

Prep Time (in minutes)

Cook Time (in minutes)

Tags
 Enter tags (e.g. chick)

Unit Types

teaspoon = tsp.	milliliter = ml.
tablespoon = tbsp.	liter = L.
cup = c.	milligram = mg.
quart = qt.	gram = g.
ounce = oz.	kilogram = kg.
pint = pt.	millimeter = mm.
gallon = gal.	centimeter = cm.
inch = in.	meter = m.
pound = lb.	celsius = C.
fahrenheit = F.	

List of Ingredients

Ingredient	Quantity	Unit	
Salt	20	mg	Delete
Ingredient	Quantity	Unit	
Egg			Delete
Ingredient	Quantity	Unit	
Pepper	20	mg	Delete

+ Add Ingredient

Figures 4.2: Add recipe

Fried Rice

Fried Rice

Ingredients

✓ 2.00 cups of Rice

✓ 2.00 pcs of Eggs

✓ 2.00 tbsp of Soy Sauce

✓ 2.00 cloves of Garlic

✓ 2.00 stalks of Green Onions

✓ 0.50 cups of Carrots

Steps

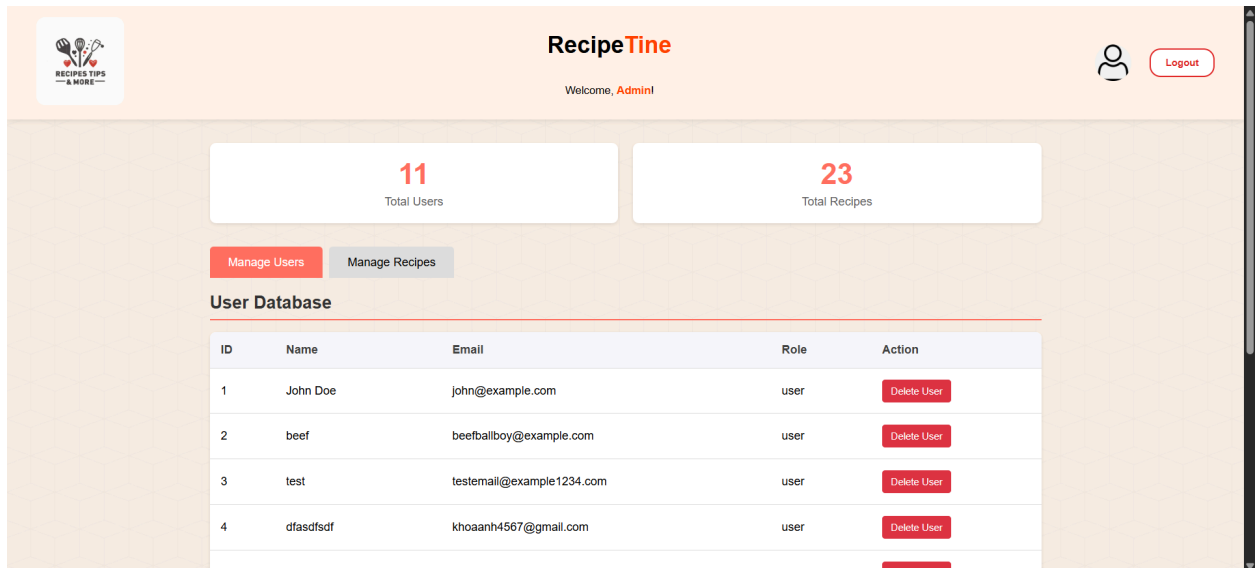
Step 1
Heat oil in a pan and sauté garlic until fragrant.
Duration: 2 minutes

Step 2
Add carrots and stir-fry for 3 minutes.
Duration: 3 minutes

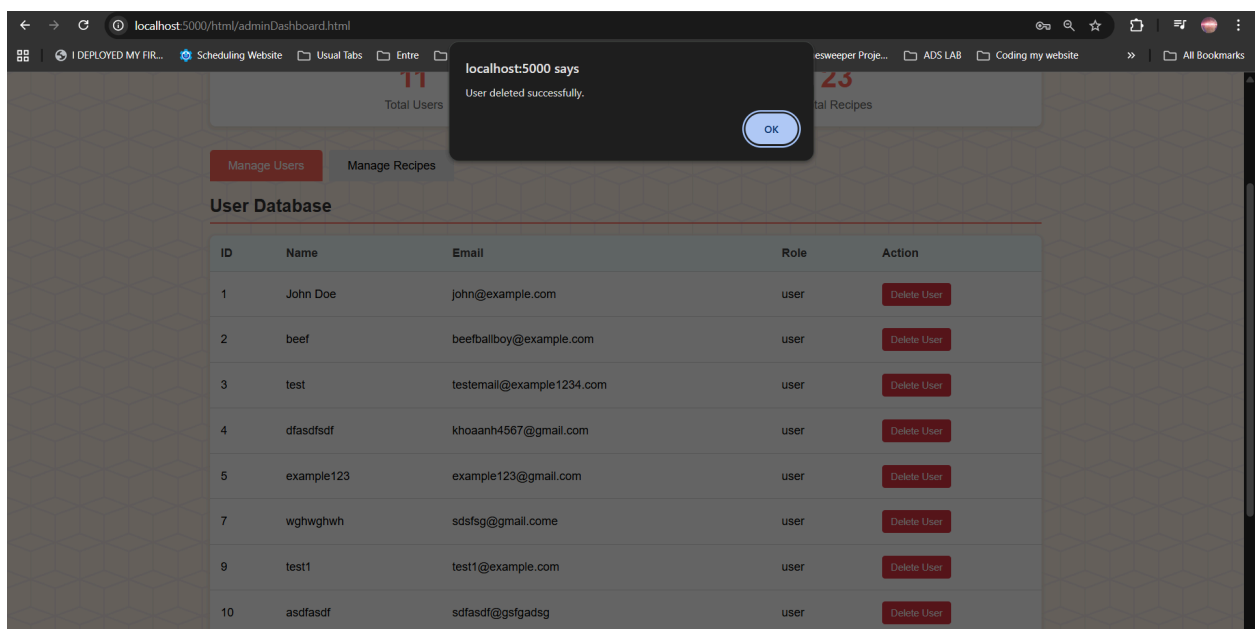
Step 3
Push vegetables aside and scramble eggs in the same pan.
Duration: 2 minutes

Step 4
Add rice, soy sauce, and green onions. Mix well and fry for 5 minutes.
Duration: 5 minutes

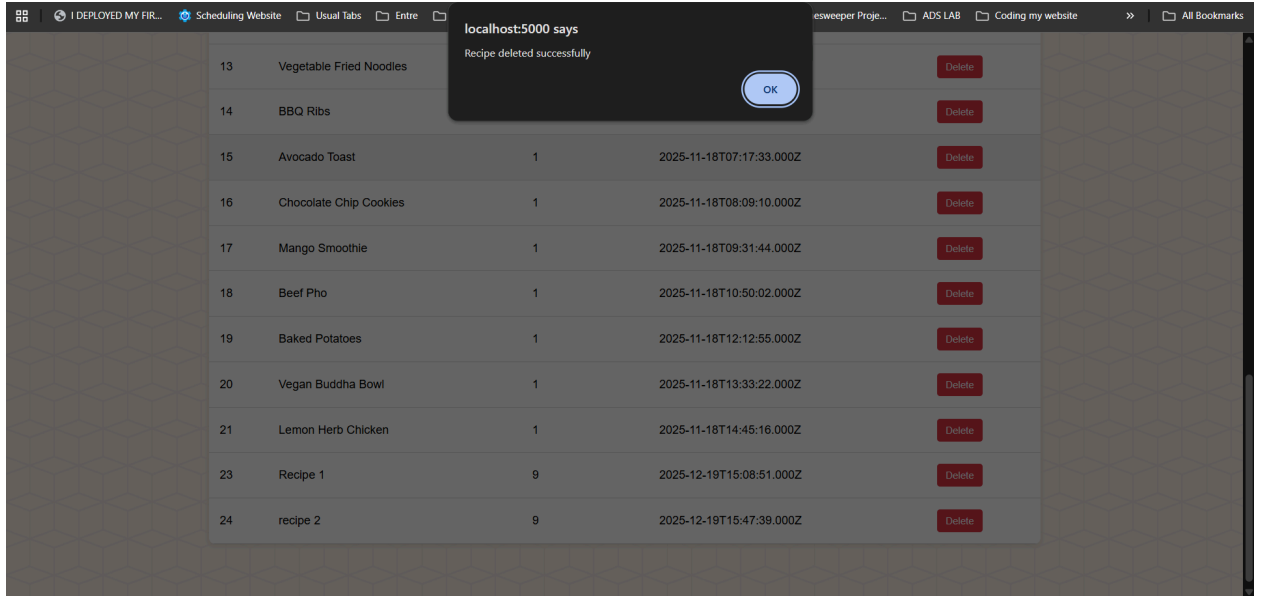
Figures 4.3: Recipe View with Steps



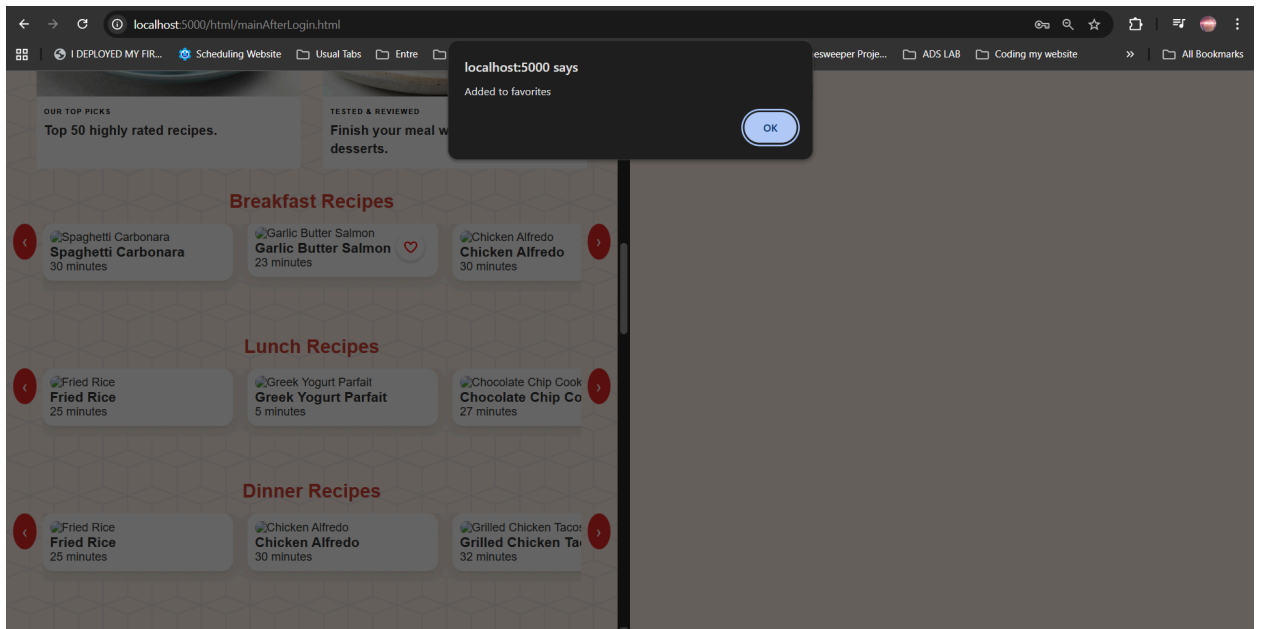
Figures 4.4: Admin Dashboard with delete user and recipe



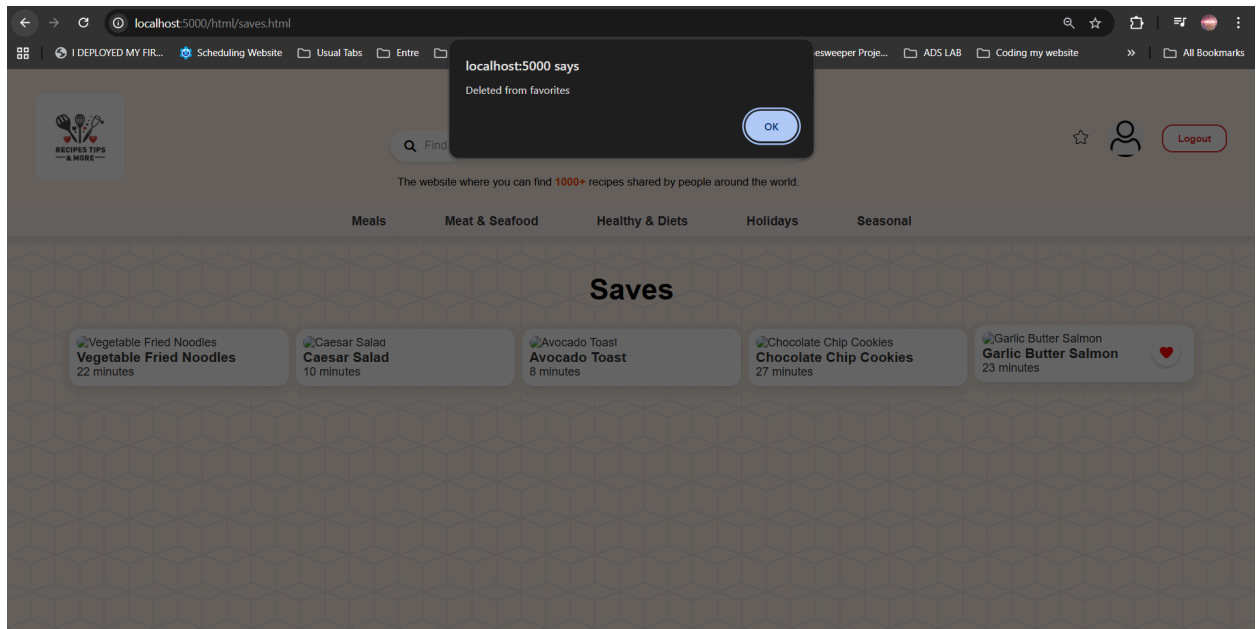
Figures 4.5: Delete user successfully



Figures 4.6: Delete recipe successfully



Figures 4.7: Add a recipe to favorites



Figures 4.8: Unfavorite a recipe

CHAPTER 3: CONCLUSION

3.1 Conclusion

Key Findings and Achievements:

- + **Main Findings:** We successfully created a website for the online recipe that allows users to see recipes of many categories. Users can register, log in, rate recipes, comment on them.
- + **Achievements:** Implementing a friendly interactive interface for users to navigate, choose food recipe tags, users are able to add their own recipe and tag them.

- Reflection on Objectives:

- + **Overall Objective Evaluation:** The project's main goals all were met. The website successfully displays services and costs, enables user registration and login, allows users to add and edit recipe, and supports recipe information and administration management.

3.2 Concluding Thoughts

The Food Recipe Website project provided an excellent opportunity to apply database management principles and web app development skills learned in the course. The project, developed collaboratively by the team, required effective communication, coordination, and understanding of theoretical concepts to translate them into a working application. Each team member gained practical experience in handling a multi-faceted project, integrating backend and frontend components, and ensuring system functionality. The knowledge acquired through this project will be beneficial for future coursework and professional work in areas of database management, web app development, and system architecture.

3.3 References

<https://github.com/HeomapTV123/Recipes.github.io>